Mohammed Faiz Nawaz

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Professional Summary

Software Engineer with **4+ years** of experience building scalable systems, automation solutions, and **AI/ML** integrations. Skilled in cloud infrastructure and security with **full-stack** development using **Java**, **Python**, **TypeScript**, **React**, **Angular**, **Node.js**, **FastAPI**, **Docker**, and **CI/CD** pipelines, over **REST** and **GraphQL** APIs. Focused on **Kubernetes**-based platforms, **AWS** cloud automation, and **microservices** and **distributed systems** at scale.

Experience

Software Engineer Jan. 2025 – Present

GrowMe.ai *Buffalo, NY*

- Engineered an **LLM powered** property search application in **TypeScript/Node.js** with prompt orchestration, structured output parsing, and fallback handling, improving search relevance by **60%** across 1,000+ daily records.
- Built and deployed 10+ secure **REST** and **GraphQL APIs** and microservices using **Node.js/Express.js** backed by **PostgreSQL**, implementing **OAuth2** authentication, Row Level Security, **Swagger** docs, and automated **Jest** testing.
- Optimized performance with **SQL indexing**, **Redis** caching, and query tuning server side alongside **React** component and state optimization client side, removing bottlenecks across the stack.
- Integrated **Elasticsearch** for full text search and log analytics with custom analyzers and relevance scoring optimizations, and explored **Neo4j** graph database patterns for modeling user preference relationships.
- Provisioned **PostgreSQL** via **Helm** on **Kubernetes** (StatefulSet + PVCs, Secrets, **NetworkPolicies**), hardening connectivity and isolation so services had a reliable transactional store without bleeding across namespaces.
- Built observability dashboards with **Prometheus**, **Grafana**, and **Datadog**, creating custom exporters and alerting on **SLIs**, cutting incident resolution from 45 minutes to under 15.
- Deployed cloud native applications on **AWS (EKS, Lambda, S3, RDS)** with **Terraform** and **AWS CDK** for infrastructure provisioning, releasing through Blue Green deployments via **ArgoCD** and **GitHub Actions**.
- Leveraged **Claude Code**, **Cursor IDE**, and **multi agent AI workflows** to accelerate development and automated test generation, working through **Agile** sprints, standups, and on-call rotation with **Jira** and **Confluence**.

Research Assistant – Software Engineer Jul. 2024 – Dec. 2024

State University of New York at Buffalo *Buffalo, NY*

- Developed **full-stack** components for the AIM Safe platform using **Python** (**FastAPI**), **React**, and **TypeScript** over **REST/JSON** APIs with non-blocking **async Python** inference pipelines on **AWS**.
- Implemented **security** controls including end-to-end encryption (**TLS 1.3**, **AES-256**), **RBAC**, audit logging, and identity management with **IAM** and **JWT** across **multi-tenant** environments.
- Built **CI/CD** pipelines with **Jenkins** and **Git**-based workflows gating **Pytest** automated tests, and used **Claude** and **GitHub Copilot** for AI-assisted development and code review.

Programmer Analyst Trainee Mar. 2023 – Jan. 2024

Cognizant Technology Solutions *India*

- Designed backend **microservices** in **Java Spring Boot** with **GraphQL** and **REST** endpoints, enforcing **OAuth2** and **JWT**-based access control across every service boundary.
- Implemented **event-driven** message processing with **RabbitMQ** and **Spring Events**, decoupling transaction logging from core flows to hold responsiveness under high load.
- Integrated **Cassandra** with **Hibernate** and **JPA** using custom **JPQL** queries for highly scalable, high-throughput record retrieval, significantly reducing data access latency.
- Designed and optimized **PostgreSQL** and **MongoDB** schemas, wrote complex **SQL** queries and stored procedures, and developed **Python ETL** workflows synchronizing data between enterprise systems.
- Conducted peer **code reviews**, ran sprint planning in **Jira**, and collaborated with **product managers**, **designers**, and QA to translate requirements into solutions, **accelerating delivery 30%**.
- Maintained **90%+ code coverage** with **JUnit** and **Mockito** across **unit and integration tests** for critical modules, setting review standards junior engineers followed on later releases.

Software Developer Feb. 2021 – Dec. 2022

Vellichor Media *India*

- Developed and maintained **Java** web application modules deployed on **Apache Tomcat** and **JBoss**, debugging stack traces and **JDBC** connection handling to resolve defects in production request paths.
- Tuned **JVM** heap settings, garbage collection, and thread pools in server.xml to resolve **thread pool exhaustion** causing intermittent 503 errors under peak load.

- Automated backups, log archival, and disk usage reporting in **Bash** and **Python** writing to **NFS** mounts, **saving 10 hours weekly** of manual effort.
- Resolved a critical disk exhaustion incident by **recovering 40 GB within two hours**, then added logrotate and cron controls so the same failure could not recur.
- Configured **Nagios** and **Splunk** monitoring across **30+ servers** with custom health checks, **cutting mean time to detection 30%** while carrying **on-call** coverage.

Projects

Project Sentinel: Autonomous Self-Healing Infrastructure | *Go, Python, gRPC, LangGraph, Docker*
- Architected a polyglot **Go** and **Python distributed system** using **gRPC** and Protocol Buffers v3 for low-latency bidirectional communication, with a StreamLogs push channel delivering anomalies without polling.
- Built a **LangGraph** agentic self-correction loop backed by **Gemini 2.5 Flash** to autonomously detect incidents, generate remediation commands, and retry failed fixes until resolution.
- Engineered a **Go**-based command policy engine validating every AI-generated command against a regex whitelist before os/exec execution, rejecting non-compliant commands with structured errors.

SmartMate: AI Agent Personal Assistant | *FastAPI, LangGraph, GPT, Google Calendar API*
- Built an **NLP**-driven agent using **GPT** for intent recognition and **LangGraph** for multi-step orchestration, automating scheduling across **Google Calendar** and messaging **APIs**.

FitFuel: AI Diet Application | *DistilGPT-2, FastAPI, Next.js, Docker, NGINX*
- Built an AI meal planner on **DistilGPT-2** served through **FastAPI** with a **Next.js** front end, Dockerized behind an **NGINX** reverse proxy for scalability and routing.
- Designed the **React** interface with form validation and **responsive** layouts, consuming the **REST API** through typed client calls with loading and error states.

TacoDB: Database Storage Engine | *C++, CMake, Docker, Linux*
- Built a database engine in **C++** implementing query planning, **query optimization**, and **B-tree indexing** over on-disk pages with slotted data pages and heap tables for dynamic record management.
- Implemented an in-memory buffer manager using **LRU** eviction that **cut disk I/O latency 60%**, and a self-balancing B-tree index reducing scan time from linear to logarithmic complexity.

TCP-Based Distributed Multi-Server Chat System | *C++, Linux CLI, Ubuntu, Bash*
- Implemented client features including user authentication, real-time message broadcasting, dynamic contact management, and presence monitoring over **TCP socket programming**.
- Developed server-side modules for message aggregation, **session state** management, and buffering messages for clients with intermittent connectivity across **distributed** servers.

Big Data Analytics: Wikipedia | *Python, Apache Spark, AWS EMR, S3, Terraform*
- Performed **ETL** on a **128 GB** dataset with **Python** and deployed a **Spark Streaming** application on **AWS EMR**, provisioning clusters in **Terraform** for reproducible workflows.

Education

State University of New York at Buffalo — Buffalo, NY
Master of Science in Computer Science — *Jan. 2024 – Jun. 2025*
- Relevant Coursework: Database Systems, Modern Concepts in Networking, Operating Systems, Analysis of Algorithms, Machine Learning, Computer Security, Data Intensive Computing.

Jawaharlal Nehru Technological University Kakinada — India
Bachelor of Technology in Electrical & Electronics Engineering — *Aug. 2018 – May 2022*
- Relevant Coursework: Computer Programming, Data Structures, Algorithms, Data Mining, Microprocessors, Cryptography, Digital Design.

Certifications

AWS Certified Solutions Architect – Associate | **AWS Certified Cloud Practitioner – Foundation** 2026

Technical Skills

Languages: TypeScript, JavaScript, Python, Java, Go, C++, SQL, HTML5, CSS3, Bash/Shell
Frontend: React, Angular, Next.js, responsive UI, component libraries, state management, Tailwind, monorepos
Backend & APIs: Node.js, Express, Spring Boot, FastAPI, NestJS, REST, GraphQL, gRPC, Swagger/OpenAPI, microservices
Databases: PostgreSQL, MySQL, MongoDB, DynamoDB, Redis, Elasticsearch, Neo4j, schema design, indexing, optimization
Cloud & Infrastructure: AWS (EKS, Lambda, S3, RDS, IAM, VPC), Kubernetes, Docker, Helm, Terraform, AWS CDK
Observability & Testing: Prometheus, Grafana, Datadog, Splunk, Jest, Pytest, Playwright, JUnit, unit and integration testing
AI & Practices: Claude Code, Cursor, Copilot, LangGraph, algorithms and data structures, Agile/Scrum, Jira, Confluence